Hydration Testing and Weigh-In Policy

Purpose and Scope

As the international federation recognised by the International Olympic Committee for all forms of cycling, including cycling esports, it is the UCI's role to regulate the sport in a way which promotes fairness and athlete safety. This policy establishes a standardized weigh-in and hydration protocol to prevent unhealthy weight-cutting practices while ensuring all riders compete on a level playing field. Rapid weight reduction through dehydration or extreme dieting is a known health risk and can distort competition fairness.

By implementing science-based guidelines (drawing from other weight-sensitive sports, and American College of Sports Medicine recommendations – ACSM -), this policy aims to protect rider welfare, discourage dangerous weight manipulation, and maintain competitive equity. In principle, all specific procedures enacted to implement this policy will be administered by a neutral medical team, with the primary focus on athlete well-being and opportunities for compliance rather than punishment.

Unhealthy Weight-Cutting: Definition and Risks

In weight-sensitive sports, "weight-cutting" typically refers to rapid loss of body mass (often 5–10% of body weight) in the days or hours before competition, usually by dehydration. Athletes may attempt to weigh in at a much lower weight than their normal weight to gain a perceived advantage or to improve power-to-weight ratio as calculated by the esports platform in use for a specific event. However, this practice carries serious health risks and performance drawbacks.

Research has shown that cutting ~5–7% of body weight through dehydration impairs cognitive function, reduces muscle power, and slows reaction time. For every kilogram of rapid weight loss, injury risk in competition may increase by about 14%. In extreme cases, dehydration has led to fatal outcomes.

Dehydration can cause overheating and cardiovascular strain. The use of diuretics or laxatives to drop water weight carries risks of electrolyte imbalance, kidney damage, and is banned in most sports, including all cycling disciplines. Severe caloric restriction or "crash" dieting can leave athletes in a state of low energy availability, which harms metabolism and endurance and, in the long term, can contribute to hormonal disorders. In addition to the emotional and physical harm caused by the risk of disordered eating and eating disorders fostered by these unhealthy behaviours.

In short, unhealthy weight-cutting endangers athletes' health and often undermines performance rather than enhancing it. The American College of Sports Medicine's consensus on weight-category sports warns that losses greater than ~3% body mass over a short period have *"substantial negative implications to health and performance."*

Rationale: The goal of this policy is to eliminate the incentive for riders to engage in rapid weight-cutting and is designed to remove any competitive advantage from dangerous short-term weight loss through dehydration, thereby protecting athlete health and promoting fair competition.

Given this, hydration testing of each athlete will be done to ensure proper hydration immediately before their official weigh-in is done.

Hydration Testing Protocol

A certified medical official uses a handheld refractometer to check an athlete's urine specific gravity (hydration status) on-site. Ensuring euhydration is critical before recording an official weight.

Method: All athletes must prove euhydration at the time of their official weigh-in by submitting to a urine specific gravity (USG) test using a calibrated handheld refractometer.

Hydration Criterion: For the purposes of cycling esports, urine specific gravity < 1.020 will be defined as a hydrated state, in line with ACSM consensus. A USG reading above or equal to 1.020 indicates the athlete is not sufficiently hydrated and will not be allowed to complete the weigh-in at that time.

Our adoption of the 1.020 standard errs on the side of safety and ensures that riders have a healthy fluid balance when establishing their official weight. Medical literature shows that urine specific gravity rises as dehydration increases, and values above ~1.020 correspond to significant body water deficit.

Testing Procedure: Hydration testing will be conducted in a controlled, private setting by a neutral medical team:

- Athletes will provide a witnessed urine sample under the supervision of a trained official of the same gender. This ensures sample validity and that no tampering occurs.

- A sample of urine will be placed on the refractometer prism, and the specific gravity reading will be taken immediately. The device typically provides a numeric readout (e.g., 1.015, 1.021, etc.). The medical official will record this value.

- **Pass or Fail:** If USG is < 1.020, the athlete is considered hydrated and clears the hydration check. If USG is greater than or equal to 1.020, the athlete has "failed" the hydration test, indicating dehydration. In a fail scenario, no weight will be recorded at that time, and the athlete must rehydrate and repeat the test. For clarity, the athlete will only be informed of the specific value of their test if they fail. Any value that passes the test will simply be shared as a pass. All specific test values will remain confidential and only a pass/fail standard will be shared as needed with relevant parties.

- For quality control, refractometers will be calibrated per manufacturer guidelines and cleaned between uses. The medical team will immediately conduct a second measurement on the same sample to double-check accuracy using the same or another refractometer.

- In the event that the two measurements are inconsistent, then a third measurement from the same sample shall be immediately taken.
    - The validity of the test will be taken by the majority of the measurements. If two measurements pass the test, then the sample is considered valid and the athlete will proceed to the weigh-in. If two measurements fail the test, then the test will fail and a retest will be offered once the athlete has an opportunity to properly and safely rehydrate.

Equipment: The recommended devices include models such as the digital ATAGO PAL-10S or MISCO PA203 handheld refractometer, which are portable and provide quick USG readings to three decimal places. At least two identical refractometers will be on hand for redundancy.

The medical team will distribute educational tips beforehand, such as avoiding diuretics, limiting caffeine/alcohol, and consuming electrolyte-containing fluids to promote hydration.

If the Hydration Test is Failed: An athlete who does not meet the < 1.020 standard will be given an opportunity and support to rehydrate. Immediately after a failed test, a member of the medical team will counsel the athlete on how to safely restore hydration levels.

For the avoidance of doubt, if an athlete cannot satisfy the testing requirements prior to the race start, they will be prohibited from starting the race.

Weigh-In Procedures

All competitors must complete the official weigh-in under the supervision of the neutral medical team. The weigh-in process is tightly coupled with the hydration checks described above – a weigh-in is not official unless the athlete passes the hydration test.

Racer Information Seminar

Prior to the weigh-in, all competitors will attend a mandatory information seminar. The session is designed to ensure that every athlete understands the health and safety principles underlying the weigh-in and hydration protocol and to promote informed, responsible participation.

The seminar will cover the following topics:

- **Definition of Unhealthy Weight-Cutting**: An explanation of what constitutes dangerous or manipulative weight-loss practices, including rapid dehydration, excessive fluid restriction, sauna use, and the use of diuretics or laxatives.

- **Health Risks of Repeated Weight Cutting**: A review of the short- and long-term consequences of engaging in weight-cutting behaviours, including impaired performance, increased injury risk, hormonal disruption, cognitive decline, and cardiovascular strain.

- **Hydration Testing Protocol**: A walkthrough of the urine-specific gravity (USG) test used to assess hydration status, including how to interpret results, why proper hydration is required before weigh-in, and how hydration status reflects an athlete's overall readiness to race safely.

- **Weigh-In Procedure Overview**: A detailed explanation of the scheduled weigh-in process, including retesting opportunities, minimum allowable race weight policies, the use of weight data in the virtual platform, and how the UCI ensures fairness and transparency.

Athletes will be provided with educational materials summarizing key points, and the medical team will be available to answer questions and offer guidance. This seminar is an essential component of the event's commitment to athlete welfare and fair competition, and attendance is required for weigh-in eligibility.

Weigh-in Conditions: Weigh-ins will be conducted using a calibrated digital scale on a hard, flat surface. Riders will weigh in with appropriate clothing to ensure accuracy (required attire: cycling shorts, jersey, and socks, no shoes). Each scale will be checked with a known calibration weight before and during the sessions. The same scales will be used for all weigh-ins to maintain consistency.

The pre-race weigh-in will be held as close as possible to the competition start time, typically between three and four hours prior to the race start. Research and expert consensus support that reducing the interval between weigh-in and competition disincentivizes rapid weight loss. By timing the final weigh-in close to the event, we reinforce the message that riders should enter the race in a hydrated, nourished state.

Medical Oversight

All aspects of weigh-ins and hydration testing will be overseen by a neutral medical team to ensure integrity, fairness, and athlete safety. This team will be assembled by the UCI and race organizers and will include qualified professionals such as sports physicians, certified athletic trainers, and sports dietitians or nutritionists. Their role is to implement the policy uniformly for all competitors and to support athletes through the process.

Neutral Supervision: The medical team members have no affiliation with any competitor or national team. Weigh-ins will be conducted with at least one official present to witness the reading. All measurements will be documented and signed off by the medical official and the athlete for transparency.

Athlete Welfare and Guidance: This policy is designed to prioritize athlete welfare at every step.

- If an athlete fails the hydration check, the medical team will immediately engage with them to devise a rehydration strategy. The athlete will be educated on how much and what to drink, and the importance of avoiding overhydration will also be explained. The medical staff may monitor vital signs if the athlete was severely dehydrated.

Compliance and Sanctions

Opportunities to Comply: Every athlete will be given ample opportunity to comply with the weigh-in requirements. Athletes will be given access to pre-race hydration test attempts, as well as the chance to address any issues that arise and ask any clarifying questions.

Sanctions for Noncompliance: Sanctions will apply only in extreme cases. Examples may include: refusing to provide a urine sample, refusal to step on the scale at the appointed times, attempting to falsify or manipulate samples, confirmed utilization of aforementioned unhealthy weight-cutting measures, deliberately ignoring direct instructions from the medical officials, and other such behaviours or actions.

If an athlete or team engages in such behaviour, the UCI Commissaire reserves the right to impose penalties, including but not limited to immediate disqualification and removal from the start list.

Our emphasis is on education, prevention, and cooperation. The medical team will brief all competitors and coaches on the requirements before the event.

Confidentiality and Fairness: The medical team will handle all personal health information with confidentiality. Results of hydration tests and weights will be communicated to the athlete in private. Every athlete is treated equally under this policy, irrespective of status or nationality. In case of any disputes, the commissaire will review the case, and a re-check may be done to ensure fairness. The guiding principle is that the health of the riders and the integrity of the competition come first.

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Conclusion

In summary, the UCI Cycling Esports Weigh-In and Hydration Policy is a comprehensive plan to eliminate dangerous weight-cutting using voluntary dehydration and promote fair competition.

This policy is intended to safeguard athletes' health. The UCI is committed to an athlete-centric approach: every rider will be given full support to meet the requirements, and punitive measures are reserved only for those who intentionally flout the rules. With this policy in place, the UCI will support riders at their healthiest and highest performance capacity, ensuring an event that is fair, competitive, and safe for all participants.

Sources:

- Burke L, Slater GJ, Matthews JJ, Langan-Evans C, Horswill CA. ACSM Expert Consensus Statement on Weight Loss in Weight-Category Sports. Curr Sports Med Rep. 2021 Apr 1;20(4):199-217.
- Sundgot-Borgen J, Meyer NL, Lohman TG, et al; How to minimise the health risks to athletes who compete in weight-sensitive sports review and position statement on behalf of the Ad Hoc Research Working Group on Body Composition, Health and Performance, under the auspices of the IOC Medical Commission British Journal of Sports Medicine 2013; 47:1012-1022.